Submission Worksheet

Submission Data

Course: IT114-450-M2025
Assignment: IT114 Module 4 Sockets Part3 Challenge
Student: Mariano R. (mr822)
Status: Submitted | **Worksheet Progress:** 100%
Potential Grade: 10.00/10.00 (100.00%)
Received Grade: 0.00/10.00 (0.00%)
Started: 6/23/2025 5:54:31 PM
Updated: 6/23/2025 8:03:46 PM
Grading Link: https://learn.ethereallab.app/assignment/v3/IT114-450-M2025/it114-module-4-sockets-part3-challenge/grading/mr822
View Link: https://learn.ethereallab.app/assignment/v3/IT114-450-M2025/it114-module-4-sockets-part3-challenge/view/mr822

Instructions

- Overview Link: https://youtu.be/_029E_aBTFo

1. Ensure you read all instructions and objectives before starting.
2. Create a new branch from `main` called `M4-Homework`
    1. `git checkout main` (ensure proper starting branch)
    2. `git pull origin main` (ensure history is up to date)
    3. `git checkout -b M4-Homework` (create and switch to branch)
3. Copy the template code from here: GitHub Repository - M4 Homework
    - It includes Sockets `Part1`, `Part2`, and `Part3`. Put all into an `M4` folder or similar if you don't have them yet (adjust `package` reference at the top if you chose a different folder name).
    - Make a copy of `Part3` and call it `Part3HW`
        - Fix the `package` and `import` references at the top of each file in this new folder (Note: you'll only be editing files in `Part3HW`)
    - Immediately record to history
        - `git add .`
        - `git commit -m "adding M4 HW baseline files"`
        - `git push origin M4-Homework`
        - Create a Pull Request from `M4-Homework` to `main` and keep it open
4. Fill out the below worksheet
    - Each Problem requires the following as you work
        - Ensure there's a comment with your UCID, date, and brief summary of how the problem was solved
        - Code solution (add/commit periodically as needed)
        - Hint: Note how `/reverse` is handled
5. Once finished, click "Submit and Export"
6. Locally add the generated PDF to a folder of your choosing inside your repository folder and move it to Github
    1. `git add .`
    2. `git commit -m "adding PDF"`

3. `git push origin M4-Homework`
4. On Github merge the pull request from `M4-Homework` to `main`

7. Upload the same PDF to Canvas
8. Sync Local
    1. `git checkout main`
    2. `git pull origin main`

Section #1: (3 pts.) Challenge 1 - Coin Flip

Progress: 100%

Task #1 (3 pts.) - Implement a Coin Flip Command

Progress: 100%

Details:

- `Client` must capture the user entry and generate a valid command per the lesson details
    - Command format must be `/flip`
- `ServerThread` must receive the data and call the correct method on `Server`
- `Server` must expose a method for the logic and send the result to everyone
    - The message must be in the format of `<who> flipped a coin and got <result>` and be from the Server
- Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Multiple screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment) from `Client`
    - Should only need to edit `processClientCommands()`
2. Snippet of relevant code showing solution (with ucid/date comment) from `ServerThread`
    - Should only need to edit `processCommand()`
3. Snippet of relevant code showing solution (with ucid/date comment) from `Server`
    - Should only need to create a new method and pass the result message to `relay()`
4. Show 5 examples of the command being seen across all terminals (2+ Clients and 1 Server)
    1. This can be captured in one screenshot if you split the terminals side by side

```
            break;
        case "flip":
            // ucid: mr822 - 06/23/2025
            server.handleFlipCommand(this);
            wasCommand = true;
            break;

        // added more cases/breaks as needed for other commands
        default:
            break;
```

```
        }
    }
```

code 1

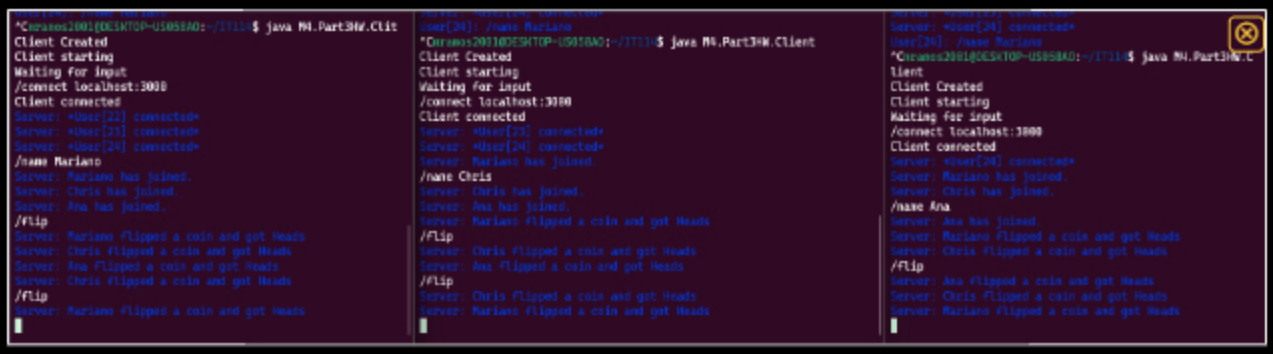
```
// ucid: mr822 - 06/23/2025
public void handleFlipCommand(ServerThread requester) {
    String result = Math.random() < 0.5 ? "Heads" : "Tails";
    String message = "Server: " + requester.getUsername() + " flipped a coin and got " + result;
    relay(null, message);
}


}
```

code 2

```
        }
        else if ("/flip".equalsIgnoreCase(text)) {
        // ucid: mr822 - 06/23/2025
        String[] commandData = { Constants.COMMAND_TRIGGER, "flip" };
        sendToServer(String.join(",", commandData));
        wasCommand = true;
        }
        return wasCommand;
    }
```

code 3



testing

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Part 2:

Progress: 100%

Details:
Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1
https://github.com/mramos822/mr822-IT114-450M4-Homework/M4/Part3HW/Client.java

URL
https://github.com/mramos822/n

URL #2
https://github.com/mramos822/mr822-IT114-450-M4-Homework/M4/Part3HW/Server.java

URL
https://github.com/mramos822/n

URL #3
https://github.com/mramos822/mr822-IT114-450-M4-Homework/M4/Part3HW/ServerThread.java

URL
https://github.com/mramos822/n

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Part 3:

Progress: 100%

Details:
Briefly explain `how` the code solves the challenge (note: this isn't the same as `what` the code does)

Your Response:

The code solves the challenge by adding a new /flip command to the client-server chat system. When a client enters /flip, it sends a specially formatted command to the server. The server detects this command and triggers a method that randomly selects either "Heads" or "Tails." The result is then broadcast to all connected clients using the relay() method. This demonstrates command parsing, server-side logic branching, and synchronized message broadcasting, fulfilling the requirement for interactive server-triggered behavior.

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Section #2: (3 pts.) Challenge 2 - Private Message

Progress: 100%

Task #1 (3 pts.) - Implement a Private Message Command

Progress: 100%

Details:

- `Client` must capture the user entry and generate a valid command per the lesson details
  - Command format must be `/pm <target id> <message>`
- `ServerThread` must receive the data and call the correct method on `Server`
- `Server` must expose a method for the logic
  - The message must be in the format of `PM from <who>: <message>` and be from the Server
  - The result must only be sent to the original sender and to the receiver/target
- Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Multiple screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment) from `Client`
    - Should only need to edit `processClientCommands()`
2. Snippet of relevant code showing solution (with ucid/date comment) from `ServerThread`
    - Should only need to edit `processCommand()`
3. Snippet of relevant code showing solution (with ucid/date comment) from `Server`
    - Should only need to create a new method and send the result message to just the sender and receiver
4. Show 3 examples of the command being seen across all terminals (3+ Clients and 1 Server)
    1. This can be captured in one screenshot if you split the terminals side by side
    2. Note: Only the sender and the receiver should see the private message (show variations across different users)

```
case "pm":
// ucid: mr822 - 06/23/2025
if (commandData.length >= 4) {
    long targetId = Long.parseLong(commandData[2].trim());
    String msg = String.join(" ", Arrays.copyOfRange(commandData, 3, commandData.length));
    server.handlePrivateMessage(this, targetId, msg);
    wasCommand = true;
}
break;
```

code 1

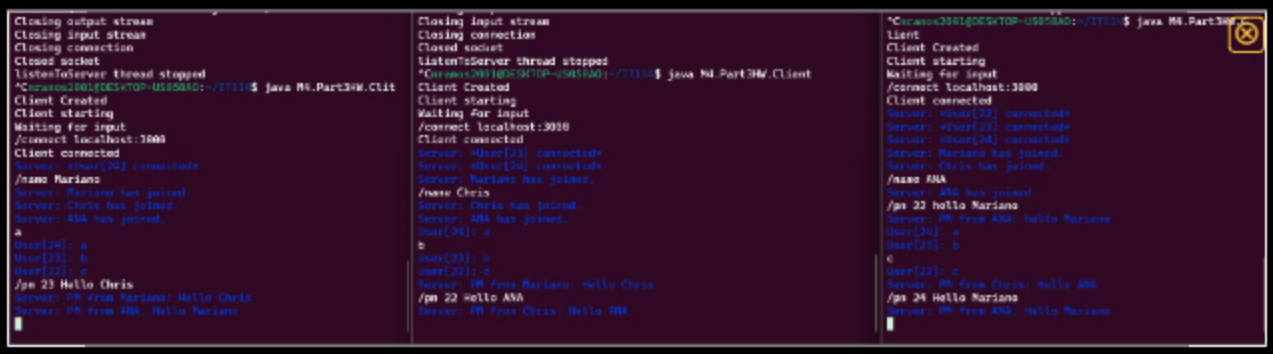
```
// ucid: mr822 - 06/23/2025
protected synchronized void handlePrivateMessage(ServerThread sender, long targetId, String msg) {
    String senderName = sender.getUsername() != null ? sender.getUsername() : "User[" + sender.getClientId() + "]";
    String formatted = "Server: PM from " + sender.getUsername() + ": " + msg;

    // Send to sender
    sender.sendToClient(formatted);

    // Send to receiver if they exist
    ServerThread receiver = connectedClients.get(targetId);
    if (receiver != null && receiver != sender) {
        receiver.sendToClient(formatted);
    }
}
```

code 2

```
    }
    else if (text.startsWith("/pm")) {
        // ucid: mr822 - 06/23/2025
        String[] split = text.trim().split(" ", 3);
        if (split.length >= 3) {
            String[] commandData = { Constants.COMMAND_TRIGGER, "pm", split[1], split[2] };
            sendToServer(String.join(",", commandData));
            wasCommand = true;
        }
    }
```

```
        return wasCommand;
    }
```

code 3



testing

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Part 2:

Progress: 100%

Details:
Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1
https://github.com/mramos822/mr822-IT114-450-M4-Homework/M4/Part3HW/ServerThread.java
URL
https://github.com/mramos822/n

URL #2
https://github.com/mramos822/mr822-IT114-450-M4-Homework/M4/Part3HW/Server.java
URL
https://github.com/mramos822/n

URL #3
https://github.com/mramos822/mr822-IT114-450-M4-Homework/M4/Part3HW/Client.java
URL
https://github.com/mramos822/n

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Part 3:

Progress: 100%

Details:
Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

The code processes the /pm command by splitting it into sender, target, and message. The client passes this to the server in a structured format. The server thread detects the "pm" command and forwards the relevant data to the Server class. The Server then constructs a private message and sends it only to the sender and the specific receiver, avoiding other clients.

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Section #3: (3 pts.) Challenge 3 - Shuffle Message

Progress: 100%

Task #1 (3 pts.) - Implement a Shuffle Message Command

Progress: 100%

Details:

- `Client` must capture the user entry and generate a valid command per the lesson details
  - Command format must be `/shuffle <message>`
- `ServerThread` must receive the data and call the correct method on `Server`
- `Server` must expose a method for the logic and send the result to everyone
  - The message must be in the format of `Shuffled from <who>: <shuffled_message>` and be from the Server
- Add code to solve the problem (add/commit as needed)

Part 1:

Progress: 100%

Details:
Multiple screenshots are expected

1. Snippet of relevant code showing solution (with ucid/date comment) from `Client`
   - Should only need to edit `processClientCommands()`
2. Snippet of relevant code showing solution (with ucid/date comment) from `ServerThread`
   - Should only need to edit `processCommand()`
3. Snippet of relevant code showing solution (with ucid/date comment) from `Server`
   - Should only need to create a new method and do similar logic to `relay()`
4. Show 3 examples of the command being seen across all terminals (2+ Clients and 1 Server)
   1. This can be captured in one screenshot if you split the terminals side by side

```
            break;
            case "shuffle":
                // ucid: mr822 - 06/23/2025
                String shuffleText = String.join(" ", Arrays.copyOfRange(commandData, 2, commandData.length));
                server.handleShuffle(this, shuffleText);
```

```
                wasCommand = true;
                break;

            // added more cases/breaks as needed for other commands
            default:
                break;
        }
```

code 1

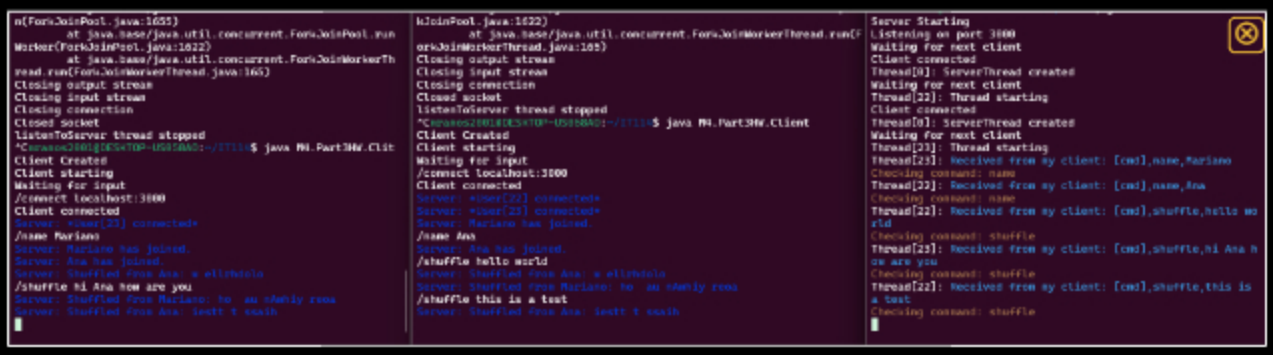
```
    // ucid: mr822 - 06/23/2025
    protected synchronized void handleShuffle(ServerThread sender, String original) {
        char[] chars = original.toCharArray();
        java.util.Collections.shuffle(java.util.Arrays.asList(chars)); // won't work
        // Fix: shuffle via list conversion
        java.util.List<Character> charList = new java.util.ArrayList<>();
        for (char c : chars) charList.add(c);
        java.util.Collections.shuffle(charList);
        StringBuilder shuffled = new StringBuilder();
        for (char c : charList) shuffled.append(c);

        String senderName = sender.getUsername() != null ? sender.getUsername() : "User[" + sender.getClientId() + "]";
        String message = "Shuffled from " + senderName + ": " + shuffled;
        relay(null, message);
    }
}
```

code 2

```
        }
        else if (text.startsWith("/shuffle")) {
            // ucid: mr822 - 06/23/2025
            String msg = text.replaceFirst("/shuffle", "").trim();
            String[] commandData = { Constants.COMMAND_TRIGGER, "shuffle", msg };
            sendToServer(String.join(",", commandData));
            wasCommand = true;
        }
```

code 3



testing

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Part 2:

Progress: 100%

Details:
Direct link to the file in the homework related branch from Github (should end in `.java`)

URL #1

https://github.com/mramos822/mr822-IT114450M4-Homework/M4/Part3HW/ServerThread.java

URL #2

https://github.com/mramos822/mr822-IT114450M4-Homework/M4/Part3HW/Server.java

URL #3

https://github.com/mramos822/mr822-IT114450M4-Homework/M4/Part3HW/Client.java

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Part 3:

Progress: 100%

Details:
Briefly explain `how` the code solves the challenges (note: this isn't the same as `what` the code does)

Your Response:

The /shuffle command solves the challenge by sending a recognizable command from the client that the server parses and processes. The server shuffles the characters of the message, formats it with the sender's name, and broadcasts it to all clients, meeting the requirement to display a randomized message from the user to everyone.

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Section #4: (1 pt.) Misc

Progress: 100%

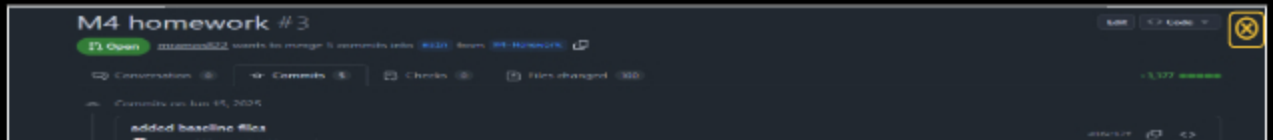
Task #1 (0.33 pts.) - Github Details

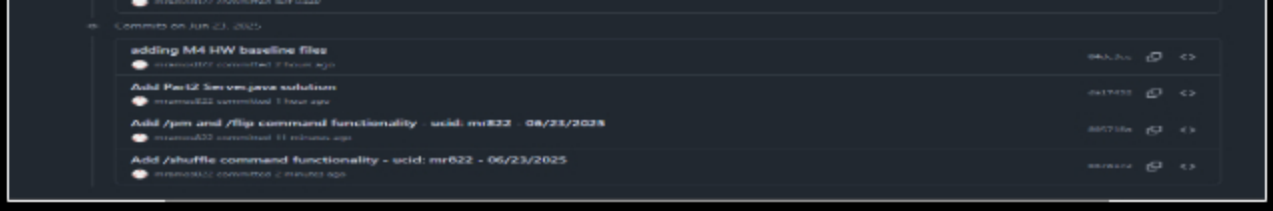
Progress: 100%

Part 1:

Progress: 100%

Details:
From the Commits tab of the Pull Request screenshot the commit history Following minimum should be present





git

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Part 2:

Progress: 100%

Details:
Include the link to the Pull Request (should end in `/pull/#`)

URL #1
https://github.com/mramos822/mr822-IT114-453/commits

URL
https://github.com/mramos822/n

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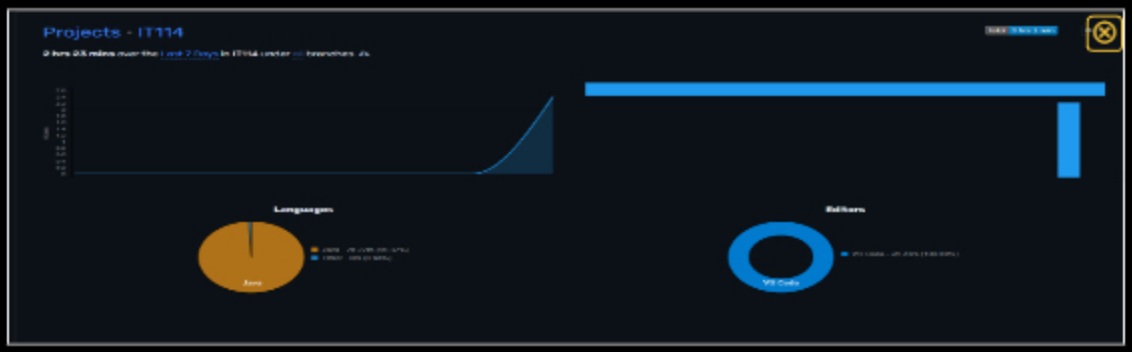
Task #2 (0.33 pts.) - WakaTime - Activity

Progress: 100%

Details:

- Visit the WakaTime.com Dashboard
- Click `Projects` and find your repository
- Capture the overall time at the top that includes the repository name
- Capture the individual time at the bottom that includes the file time
- Note: The duration isn't relevant for the grade and the visual graphs aren't necessary



waka 1

58 mins	M4/Part3HW/ServerThread.java	2 hrs 23 mins	M4-Homework
44 mins	M4/Part3HW/Client.java		
31 mins	M4/Part3HW/Server.java		
5 mins	M4/Part1/Client.java		
1 min	M4/Part3HW/TextFX.java		
58 secs	../3-Homework/.gitignore:Zone.Identifier		
46 secs	M4/Part3HW/Constants.java		

waka 2

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Task #3 (0.33 pts.) - Reflection

Progress: 100%

Task #1 (0.33 pts.) - What did you learn?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

I learned how to handle client-server communication using sockets in Java, how to structure commands like /flip, /pm, and /shuffle, and how to route them through different layers of the application.

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Task #2 (0.33 pts.) - What was the easiest part of the assignment?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

The easiest part of the assignment was adding new commands in the client and routing them through the server thread. Once I understood the structure from existing commands like /reverse, it was straightforward to follow the same pattern for /flip, /pm, and /shuffle. The command parsing and sending data with sendToServer() became very intuitive after the first one.

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Task #3 (0.33 pts.) - What was the hardest part of the assignment?

Progress: 100%

Details:
Briefly answer the question (at least a few decent sentences)

Your Response:

The hardest part of the assignment was understanding the problem itself at the beginning. It took time to figure out exactly how each command was supposed to work, what the expected output should be, and how the client and server were supposed to interact. Once I understood the structure and flow, the implementation became much easier.

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